BrightSpring

An online tutoring website where learners browse subjects and meet tutors, built with Claude Code

Project folder: https://drive.google.com/drive/folders/1WtGm9YPRXGawqDxVhmzRf6CCH9XBZDs1
Open **README.md** in the folder first. It lists where every file below lives.

The approach

Alongside the site, I set up the tooling that keeps it consistent as it grows: a design system, validated content, an automated design reviewer, a reusable skill, and project memory that survives a brand-new session. My rule throughout: define the rules first, automate the checks, and prove every tool works by making it fail on purpose.

1 Design system: one source of truth for how the site looks

WHAT I DID	HOW I CHECKED IT	WHAT I GOT
Collected 4 reference screenshots, then had Claude turn them into a style guide, design tokens and 6 component specs (navigation, buttons, subject card, tutor card, badges, form fields).	Traced one value end to end: rule R.1 (cards use a 10px radius), to the token --bs-radius-md, to the subject card spec, to *rounded-md* in SubjectCard.tsx.	The chain holds with no gaps. Open questions are logged in the style guide's gap table rather than hidden.

2 Structured content: data that is checked, then rendered

WHAT I DID	HOW I CHECKED IT	WHAT I GOT
Wrote the rules and JSON schemas first: unique ids, required fields, allowed categories, every tutor's subjects must exist, every subject needs a tutor. Then created 8 subjects and 6 tutor profiles, and rendered the page directly from those files.	Ran the validator on the real data, then on 4 deliberately broken copies (wrong category, unknown subject id, duplicate id, subject with no tutor). A script also compared the running page to the data.	The real data passes. All 4 broken copies failed with a clear message. The live page matches the data: 8 of 8 subjects and 6 of 6 tutors.

3 Design reviewer: a subagent that audits the code

WHAT I DID	HOW I CHECKED IT	WHAT I GOT
Built *design-system-reviewer*, a read-only subagent that checks components against the style guide, tokens and specs. It may only report a violation if it can cite the exact rule.	Planted 3 violations in the subject card: a corner radius, a heading colour outside the palette, and a custom font size. Ran the agent, then restored the file and confirmed it by checksum.	Caught 3 of 3, each tied to its rule (R.1; C.4 and A.1; T.2 and T.4). The clean code reports 0 violations. Its first run also found a real gap, a missing *overflow-hidden*. I fixed it and reran the test from a clean baseline.

4 Planning and project memory: context that survives a new session

WHAT I DID	HOW I CHECKED IT	WHAT I GOT
Wrote a planning prompt for the four pages and a CLAUDE.md covering purpose, pages, constraints, commands, file map and working rules.	Opened a brand-new session and asked one short question with no background. Then checked every statement in the reply against the project files.	It named the project, pages, design and data files, constraints and the right next step, with nothing re-explained. That session used 51.6k tokens of context.

5 Custom skill: rough notes in, valid data out

WHAT I DID	HOW I CHECKED IT	WHAT I GOT
Built *tutoring-content-formatter*. It turns rough subject notes into schema-valid JSON, flags duplicate ids, and asks for missing facts instead of inventing them.	Ran it on 5 messy notes (mixed casing, shorthand, inconsistent levels), then ran the real validator on the output. Also forced an invalid record through as a control.	4 notes became valid records and pass. The 5th (SAT prep) was held back because levels and topics were missing. The invalid control failed validation as expected.

Effort

Session	Work	Context size (/context)
1	Design system, content, validator, Home page, reviewer agent	220.9k of 1.0M (22%)
2	Reviewer test, custom skill, CLAUDE.md, planning prompt	143.6k of 1.0M (14%)
3	New-session test of CLAUDE.md	51.6k of 1.0M (5%)
Total	**3 sessions**	**416.1k tokens**

Figures are the context-window size that Claude Code reported for each session. It does not report a billed total, so none is estimated.